

# **Anna University**

## **REGULAR EXAMINATION ? 2025**

Artificial Intelligence ? Regular Examination, 2025

Subject Code: CS503 Semester: 5 Regulation: R23

Time: 3 Hours Max. Marks: 70

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Note: This is a demo question paper generated for Backlog Buddy.

Answer all questions. Assume suitable data wherever necessary.  
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### **SECTION A (Unit 1: Uninformed and Informed Search)**

1. Solve a problem related to Uninformed and Informed Search. Show all the steps clearly. [12 Marks]
2. Define Uninformed and Informed Search. Explain its significance in Artificial Intelligence with a suitable example. [12 Marks]
3. Define Uninformed and Informed Search. Explain its significance in Artificial Intelligence with a suitable example. [12 Marks]

### **SECTION B (Unit 2: Knowledge Representation)**

4. Write short notes on Knowledge Representation. [12 Marks]
5. Discuss the engineering applications of Knowledge Representation. [12 Marks]
6. Define Knowledge Representation. Explain its significance in Artificial Intelligence with a suitable example. [12 Marks]

### **SECTION C (Unit 3: Logic and Inference)**

7. Discuss the engineering applications of Logic and Inference. [12 Marks]
8. Explain Logic and Inference with a neat diagram/sketch and a suitable example. [8 Marks]
9. Define Logic and Inference. Explain its significance in Artificial Intelligence with a suitable example. [12 Marks]

### **SECTION D (Unit 4: Uncertainty and Bayesian Networks)**

10. Discuss the engineering applications of Uncertainty and Bayesian Networks. [12 Marks]
11. Solve a problem related to Uncertainty and Bayesian Networks. Show all the steps clearly. [12 Marks]
12. List the advantages and limitations of Uncertainty and Bayesian Networks. Explain any two in detail. [12 Marks]

### **SECTION E (Unit 5: Expert Systems and Applications)**

13. Discuss the engineering applications of Expert Systems and Applications. [12 Marks]
14. Explain Expert Systems and Applications with a neat diagram/sketch and a suitable

**example. [12 Marks]**

15. How is Expert Systems and Applications implemented? Give an algorithm / procedure and explain. [12 Marks]

\*\*\* End of Question Paper \*\*\*